

Secure Infrastructure & Network Systems

Professional Case Study

Overview

Designed and implemented a secure infrastructure combining Linux and Windows Server environments. Focused on secure communications, network configuration, and system administration.

System Architecture

Client	Secure Network	Server
SSH / SCP	Firewall (UFW)	Linux Server
FTPS Client	IPv6 Network	Apache + HTTPS
VoIP Client	Routing & Rules	Asterisk Server

Key Features

- 1 Secureremote access using SSH with custom ports
- 2 Encrypted file transfer via FTPS and SCP
- 3 HTTPS configuration using SSL/TLS certificates
- 4 Firewall rules with controlled access (UFW)
- 5 IPv6 network configuration and testing
- 6 User management and ACL permissions in Linux
- 7 VoIP communication system using Asterisk

Technologies Used

- 1 Linux(UbuntuServer)
- 2 Windows Server
- 3 Apache
- 4 OpenSSH
- 5 UFW Firewall
- 6 Asterisk VoIP
- 7 SSL/TLS & PKI

Results

Designed and deployed a secure and fully functional infrastructure, demonstrating real-world networking, system administration, and security practices. Enabled reliable communication, encrypted data transfer, and controlled access using firewall rules, secure protocols, and certificate-based authentication.

Detailed documentation available upon request (Spanish).



CASE STUDY

Secure Network & Server Infrastructure

Linux & Windows Server | Security • Networking • Services



6+
Services



100%
Secure



End-to-End
Implementation

1. OVERVIEW

Designed and implemented a secure client-server infrastructure using Linux and Windows Server. The solution integrates multiple network services and security mechanisms including SSH, FTPS, HTTPS, IPv6, Firewall rules and VoIP (Asterisk), ensuring confidentiality, integrity and availability of data and communications.



Security
First



Reliable
Network



Encrypted
Communication



VoIP
Ready

2. TECHNOLOGIES



Linux
(Ubuntu)



Windows
Server



Apache
HTTP Server



Asterisk
(VoIP)

OpenSSH

OpenS
(SSH)



FTPS
(FileZilla)



UFW
Firewall



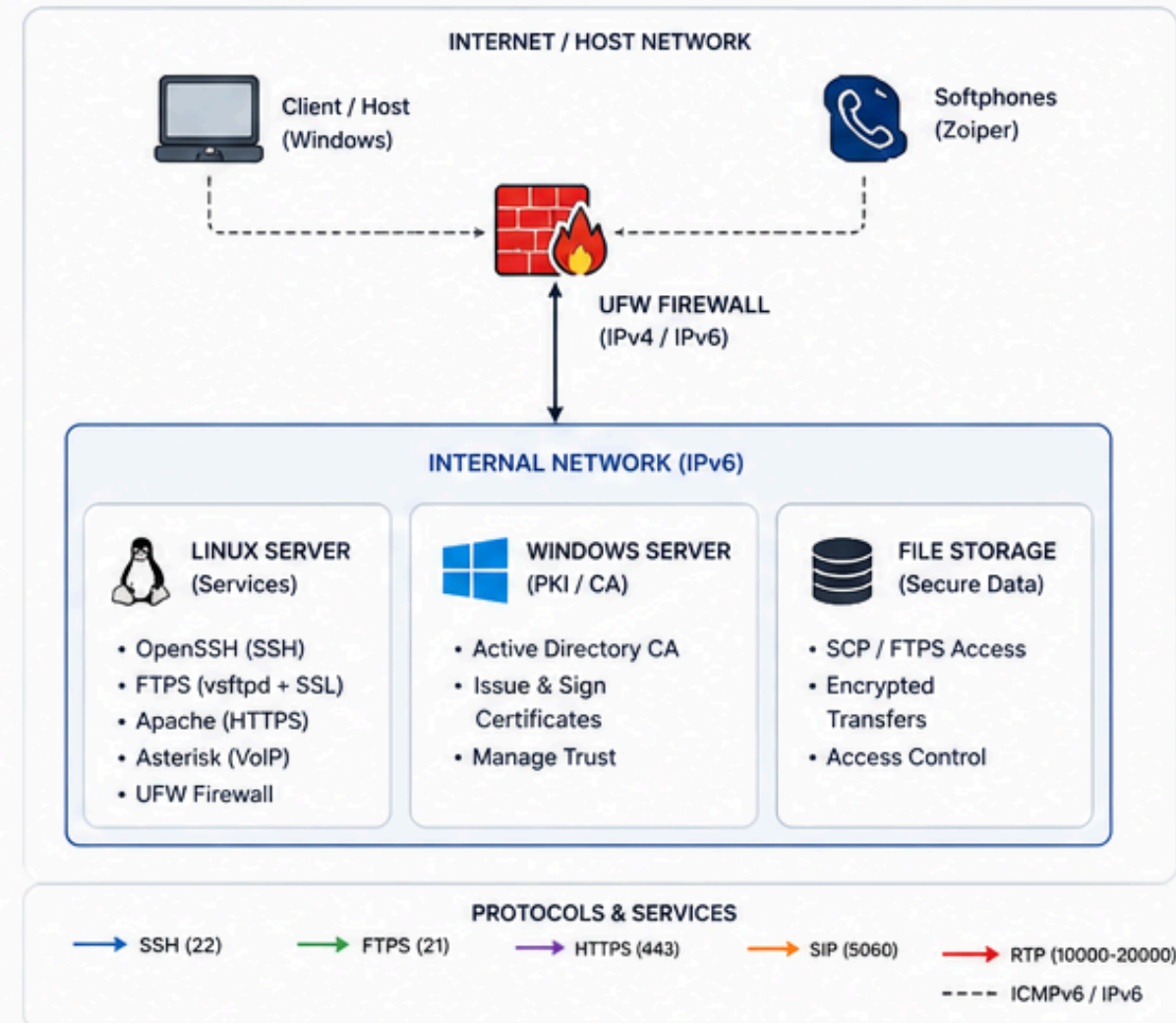
IPv6
Networking

3. OBJECTIVE

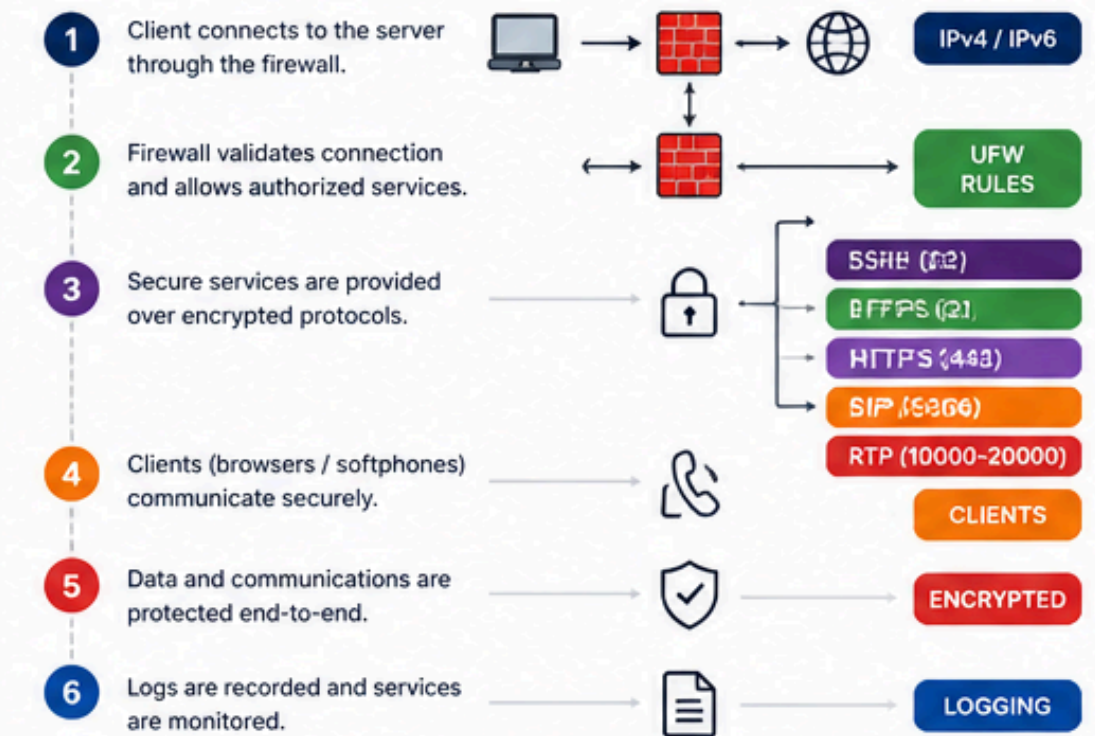
- Implement secure file transfer and remote access.
- Deploy HTTPS with trusted certificates using a custom CA (PKI).
- Configure IPv6 network for internal communication.
- Implement VoIP service for internal calls.
- Apply firewall rules to protect all services.
- Validate all services with real client connectivity.



4. INFRASTRUCTURE ARCHITECTURE



6. SECURITY & NETWORK FLOW



7. RESULTS



100%
Services
Operational



Secure
Encrypted
Communication



IPv6
Fully Functional
Network

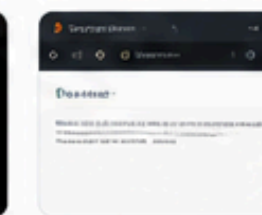


VoIP
Internal Calls
Successful

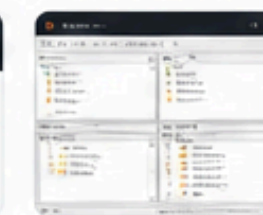
8. EVIDENCE (SCREENSHOTS)



IPv6 Connectivity
Test



HTTPS Web Server
(web1.spl-project.local)



FTPS Connection
(FileZilla)



VoIP Call
(Zoiper)



Built with **Security in Mind**
Reliable. Scalable. Professional.



Key Takeaway

This project demonstrates the ability to design, secure and deploy real-world infrastructure services.



Author

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